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Pump housing layout for brake system

Abstract

According to at least one embodiment, the present disclosure provides a pump system for a brake system, comprising: an oil reservoir; a pump housing having a suction port and a discharge port formed therein; a motor disposed at a side of the pump housing; an electronic control unit disposed at another side of the pump housing; a pump installed in the pump housing and having a suction channel, which is connected with the suction port, and a discharge port, which is connected with the discharge port, therein; a shaft coupled to the motor to rotate and having a rotating body configured to operate the pump; and a pump-reservoir pipeline configured to connect the suction channel to the oil reservoir.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2021-0067080, filed on May 25, 2021, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field
- (2) The present disclosure relates to a pump housing layout for a brake system. In more detail, the present disclosure relates to a pump housing layout for a brake system employing a BLAC (Brushless Alternating Current) motor.
2. Discussion of Related Art
- (3) Description of this section only provides the background information of the present disclosure

without configuring the related art.

(4) A brake system of a vehicle includes a motor to secure a braking force that is required to stop or decelerate a vehicle. A DC (Direct Current) motor is mounted on several brake devices to implement ABS (Anti-lock Brake System) or an ESC (Electronic stability control).

(5) However, a DC motor is not suitable for performing additional functions that should be always performed because operation noise due to an RPM change is large. There is an alternative plan of using a BLAC (Brushless Alternating Current) motor of which noise is small instead of a DC motor. However, when a BLAC motor is used, there is a problem in that it is possible to additionally secure an installation space in a vehicle in order to install a device for sensing the rotational position of a shaft coupled to the motor to rotate.

(6) FIG. 1 is a cross-sectional view showing an ABS module equipped with a BLAC motor in the related art.

(7) Referring to FIG. 1, a pump housing layout that prevents an increase in installation space of a brake device by coupling a sensed component 150 to an end of a shaft 110 extending to an electronic control unit 130 and installing a sensor 170 for sensing rotation of the sensed component 150 at a position facing the sensed component 150 on the electronic control unit 130 has been proposed in order to solve this problem.

(8) However, such a pump housing layout has another problem that when a working liquid leaks from a pump operating while being in contact with a rotary member coupled to the shaft, the leaking working fluid flows to the electronic control unit due to the extending shaft and causes mechanical breakdown.

SUMMARY

(9) According to at least one embodiment, the present disclosure provides a pump housing layout for a brake system, comprising: an oil reservoir; a pump housing having a suction port and a discharge port formed therein; a motor disposed at a side of the pump housing; an electronic control unit disposed at another side of the pump housing; a pump installed in the pump housing and having a suction channel, which is connected with the suction port, and a discharge port, which is connected with the discharge port, therein; a shaft coupled to the motor to rotate and having a rotating body configured to operate the pump; and a pump-reservoir pipeline configured to connect the suction channel to the oil reservoir.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view showing an ABS module equipped with a BLAC motor in the related art.

(2) FIG. 2 is a cross-sectional view showing an oil-reservoir pipeline of a pump housing layout for a brake system according to an embodiment of the present disclosure.

(3) FIG. 3 is a cross-sectional view showing an oil-reservoir pipeline of a pump housing layout for a brake system according to another embodiment of the present disclosure.

(4) FIG. 4 is a cross-sectional view of the pump housing layout for a brake system according to an embodiment of the present disclosure.

(5) FIG. 5 is a perspective view showing a cover of the pump housing layout for a brake system according to an embodiment of the present disclosure.

(6) FIG. 6 is a cross-sectional view showing the cover of the pump housing layout for a brake system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(7) Hereinafter, some exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the following description, like reference numerals

preferably designate like elements, although the elements are shown in different drawings. Further, in the following description of some embodiments, a detailed description of known functions and configurations incorporated therein will be omitted for the purpose of clarity and for brevity.

(8) Additionally, various terms such as first, second, A, B, (a), (b), etc., are used solely to differentiate one component from the other but not to imply or suggest the substances, order, or sequence of the components. Throughout this specification, when a part 'includes' or 'comprises' a component, the part is meant to further include other components, not to exclude thereof unless specifically stated to the contrary. The terms such as 'unit,' 'module,' and the like refer to one or more units for processing at least one function or operation, which may be implemented by hardware, software, or a combination thereof.

(9) Accordingly, the present disclosure has been made to solve the problems described above and a main object is to provide a pump housing layout (or a pump system) for a brake system that prevents a working fluid from leaking to an electronic control unit from a pump housing by connecting a suction channel of a pump to an oil reservoir.

(10) The objects of the present disclosure are not limited to the objects described above and other objects will be clearly understood by those skilled in the art from the following description.

(11) A pump housing layout for a brake system according to an embodiment of the present disclosure includes all or some of an oil reservoir **210**, a pump housing **220**, a motor **230**, an electronic control unit **411**, a shaft **250**, a pump **240**, a pump-reservoir pipeline **270**, a sensed component **260**, a sensor **320**, and a cover **500**.

(12) The oil reservoir **210**, which is a storage unit configured to store or supply a working fluid of the brake system, may be connected with a master cylinder (not shown), a pedal simulator (not shown), and a pump **240**.

(13) Referring to FIG. 2, the oil reservoir **210** according to an embodiment of the present disclosure is connected with a suction channel **241** of the pump **240** by the pump-reservoir pipeline **270**. Since the suction channel **241** of the pump **240** and the oil reservoir **210** are connected such that fluid can flow, a working fluid pressurized by operation of the pump **240** can be guided to the oil reservoir **210**. Accordingly, there is an effect that it is possible to prevent a working fluid from leaking from the pump **240** and flowing to the shaft **250** due to high pressure. The pump-reservoir pipeline **270** may be a hose made of a flexible material in order not to be damaged or loosened by hitting against other parts of a vehicle.

(14) The motor **230** is disposed at a side of the pump housing **220** and an ECU housing **410** is disposed at another side of the pump housing **220**. An electronic control unit **411** and a portion of a solenoid valve are accommodated in the ECU housing **410**.

(15) A suction port **221** and a discharge port **223** are formed in the pump housing **220** according to an embodiment of the present disclosure. The suction port **221** and the discharge port **223** are connected with a suction channel **241** and a discharge channel **242** of the pump **240**, respectively. The pump **240** is installed in the pump housing **220** and the pump **240** includes pistons **243** at both sides which are reciprocated by the motor **230**, etc. When the suction channel **241** formed in the piston **243** is selectively connected with the suction port **221** formed in the pump housing **220** by reciprocation of the piston **243** installed in the pump housing **220**, oil flowing from an accumulator (not shown) can flow into the suction channel **241** through the suction port **221**, can be pressed by operation of the piston **243**, and then can be discharged to the discharge port **223** formed in the pump housing **220**.

(16) The motor **230** is disposed at a side of the pump housing **220**. The motor **230** provides rotation power for operating the pump **240** by being operated by power. The motor **230** according to an embodiment of the present disclosure may be a BLAC (Brushless Alternating Current) motor. The BLAC motor is a brushless motor having a sinusoidal back electromotive force. The BLAC motor includes a rotor and a stator installed around the rotor without a brush and a commutator, and rotates the rotor by changing the direction of a current in accordance with a rotational angle. Such a

BLAC motor has an advantage that torque can be directly controlled and noise is small in comparison to a BLDC (Brushless Direct Current) motor. The shaft **250** may be coupled to the rotor of the BLAC motor to rotate together. However, a BLAC motor requires the sensed component **260**, which rotates with the shaft **250**, and the sensor **320** that senses a position change of the sensed component **260** in order to sense the rotation position thereof.

(17) The electronic control unit **411** is disposed at another side of the pump housing **220**. The electronic control unit **411** is a control circuit board printed with electronic devices for controlling rotation of the motor **230**, opening/closing of the solenoid valve, etc. The sensor **320** that senses rotation of the sensed component **260** mounted on the shaft **250** is coupled at a position facing the shaft **250** on the electronic control unit **411**.

(18) The shaft **250** is coupled to the motor **230** to rotate. In detail, the end of the shaft **250** which faces the motor **230** is coupled to the rotor of the motor **230** to be able to rotate. A rotating body **251** that operates the pump **240** is coupled to the shaft **250**.

(19) When the piston pump **240** that is driven by the reciprocating piston **243** is installed, the rotor **251** may be an eccentric bearing **251** rotating with an area center at a position biased from the rotational axis of the shaft **250**. The eccentric bearing **251** is configured to come in contact with and press a pair of pistons **243**, which are in contact with both sides of the eccentric bearing **251**, respectively, in accordance with a rotational position while rotating with the shaft **250**. The piston **243** installed in a direction perpendicular to the rotational axis of the shaft **250** pressurizes and sends a working fluid while reciprocating by the eccentric bearing **251**.

(20) When a gear type pump **240** that pressurizes a working fluid using rotation of gears engaged with each other is installed, the rotating body **251** may be a gear having teeth engaged with a driven gear **330** of the pump **240**.

(21) The shaft **250** according to an embodiment of the present disclosure extends toward the electronic control unit **411**. The shaft **250** may be a single body coupled to the rotor of the motor **230** and axially elongated. The end of the shaft **250** which faces the electronic control unit **411** is disposed sufficiently close to the electronic control unit **411** so that magnetism of the sensed component **260** mounted at an end of the shaft **250** can be transmitted to the sensor **320**.

(22) The pump **240** is installed in the pump housing **220**, and the suction channel **241** connected with the suction port **221** and the discharge channel **242** connected with the discharge port **223**. The suction port **221** and suction channel **241**, the discharge port **223** and discharge channel **242**, and the suction channel **241** and discharge channel **242** are each configured to be selectively connected by hydraulic pressure. The suction channel **241** and the discharge channel **242** may be thin cylindrical pipelines, but are not limited thereto and may be chambers having a large cross-sectional area. The pump **240** is disposed in a pair at the left and right sides of the rotational axis of the shaft **250** and may be configured such that the piston **243** included therein moves inside and outside every time the rotating body **251** coupled to the shaft **250** is rotated by 180 degrees.

(23) Further, according to an embodiment of the present disclosure, a plurality of pumps **240** may be provided to be spaced apart from each other in a direction perpendicular to the axial direction of the shaft **250**.

(24) The pump **240** has a counterpart that reciprocates or rotates while being in contact with the rotating body **251** installed on the axis of the shaft **250**. The counterpart may be disposed in a direction perpendicular to the rotational axis of the shaft **250**. In this embodiment, the counterpart may be the piston **243** reciprocating while being in contact with the eccentric bearing **251** rotating with the shaft **250** when the pump **240** is a piston pump **240**. The pump **240** may be installed in a direction perpendicular to the rotational axial direction of the shaft **250**.

(25) In the piston pump **240**, referring to FIG. 2, a spherical body **246** is installed in an opening on the movement axial line of the suction channel **241** formed at the piston **243**. The spherical body **246** is supported by a spring **245** to come in close contact with a retainer **244** mounted at the front end of the piston **243**. The piston pump **240** is configured such that the suction channel **241** can be

opened/closed by elasticity of the spring **245** which is applied to the spherical body **246** and the pressure of oil passing through the suction channel **241**.

(26) In the pump **240**, the maximum pressure may be higher at the discharge channel **242** through which a working fluid is compressed and guided at high pressure by operation of the pump **240** than at the suction channel **241** that receives a working fluid from the suction port **221**.

Accordingly, leakage of a working fluid in the pump more easily occurs usually near the discharge channel **242** and the discharge port **223** of the pump **240**.

(27) The pump housing layout for a brake system according to an embodiment of the present disclosure uses the piston pump **240** that is operated by the piston **243** configured to reciprocate while being in contact with the rotating body **251** when the rotating body **251** is rotated. Further, the suction channel **241** is formed at a position closer than the discharge channel **242** from the rotational axis of the shaft **250** in the radial direction of the shaft **250**.

(28) The pump housing layout for a brake system according to an embodiment of the present disclosure uses the piston pump **240** that is operated by the piston **243** configured to reciprocate while being in contact with the rotating body **251** when the rotating body **251** is rotated. In this case, the suction channel **241** of the pump **240** is formed at a position closer than the discharge channel **242** from the sensed component **260** in the rotational axial direction of the shaft **250**. For example, as in FIG. 2, the suction channel **241** may be formed to be hollow toward the electronic control unit **411** and the discharge channel **242** may be formed to be hollow toward the electronic control unit **411** at a position further spaced than the suction channel **241** in the radial direction of the shaft **250**. The suction port **221** and the discharge port **223** of the pump housing **220** which are connected with the suction channel **241** and the discharge channel **242** are also formed at positions corresponding thereto.

(29) The pump housing layout for a brake system according to an embodiment of the present disclosure has an effect of preventing a leaking working fluid from flowing into the electronic control unit **411** along the extending shaft **250** by disposing the discharge channel **242**, at which the possibility of leakage of a working fluid is high, to be spaced from the shaft **250** further than the suction channel **241**.

(30) The pump **240** is not limited to the piston pump **240**. Referring to FIG. 3, for example, the pump **240** according to an embodiment of the present disclosure may be a gear pump **300** that sends a working fluid using a driving gear **310** coupled to the shaft **250** to rotate and a driven gear **330** rotating in mesh with the driving gear **310**. The suction port **221** and the discharge port **242** may be disposed on circumferential contact surfaces having the rotational axes of the driving gear **310** and the driven gear **330** as centers, respectively. An inlet port and an outlet port of the gear pump may be understood as an inlet-side chamber and an outlet-side chamber, respectively, in the present disclosure. In FIG. 3, the forming direction of the oil-reservoir is freely shown and is not limited to the direction shown in the figure.

(31) The suction port **221** is disposed at a position closer to the sensed component **260** than the discharge port **223**. In detail, the suction port **221** may be formed at a position closer than the discharge port **223** from the sensed component **260** in the rotational axial direction of the shaft **250**. There is an effect of preventing a leaking working fluid from flowing into the electronic control unit **411** by the extending shaft **250** by disposing the discharge port **223**, at which the possibility of leakage of a working fluid is high, to be spaced from the shaft **250** further than the suction port **221**.

(32) Accordingly, in the layout of the pump housing **220** of a brake system according to an embodiment of the present disclosure, the suction channel **241** is connected with the oil reservoir **210** so that a high-pressure working fluid can be guided to the oil reservoir **210** without leaking out of the pump housing **220**. Further, there is an effect of preventing oil from leaking to the electronic control unit **411** in multiple steps by disposing the discharge channel **242** to be spaced from the sensed component **260** or the shaft **250** further than the suction channel **241**.

- (33) According to an embodiment of the present disclosure, a plurality of pumps **240** may be installed to be spaced apart from each other in the rotational axial direction of the shaft **250**. It may be possible to apply connection structure with the oil reservoir **210**, the disposition of channels, etc. described above to only one or a pair of pumps **240** positioned close to the sensed component **260** of the shaft **250** of the plurality of pumps **240**.
- (34) The shaft **250** of the pump housing layout according to an embodiment of the present disclosure extends toward the electronic control unit **411** and the sensed component **260** is mounted at the end of the shaft **250** which faces the electronic control unit **411**. Further, a rotation sensor **320** that senses rotation of the sensed component **260** is installed at a position facing the sensed component **260** on the electronic control unit **411**. Accordingly, it is not required to additionally secure a space in order to install a device for monitoring rotation of the shaft **250** using the BLAC motor **230**.
- (35) The sensed component **260** is made of a magnetic material. The center of the rotation sensor **320** determines the rotational position of the motor **230** on the basis of a change in magnetic flux that is sensed in accordance with rotation of the sensed component **260** having magnetism.
- (36) Referring to FIGS. **4** to **6**, the electronic control unit **411** according to an embodiment of the present disclosure is separated from the inside of the pump housing **220** and disposed outside the pump housing **220**. This is for preventing a working fluid from leaking to the electronic control unit **411** from the inside of the pump housing **220**.
- (37) The pump housing **220** according to an embodiment of the present disclosure has an inner space **225** therein for accommodating the shaft **250**. The inner space **225** is not formed in the electronic control unit **411**. That is, the inner space **225** is formed with a predetermined thickness remaining at a position facing the sensed component **260** on the wall facing the electronic control unit **411** of the pump housing **220**. In this case, the pump housing **220** may be made of a material having high permeability at the entire portion or a portion facing the sensed component **260** so that the magnetism of the sensed component **260** can be sensed by the rotation sensor **320** installed at the electronic control unit **411**. The surface facing the sensed component **260** is for preventing leakage of oil by separating the pump housing **220** and the electronic control unit **411**, so it can be formed to be sufficiently thin.
- (38) Unlike the above description, in the pump housing **220** according to an embodiment of the present disclosure, an inner space **225** may be formed in the electronic control unit **411**. Further, a cover **500** that covers the inner space **225** at the electronic control unit **411** may be provided to separate the electronic control unit **411** and the inside of the pump housing **220**.
- (39) A flange portion **510** coupled to the outer surface of the pump housing **220** is formed at the cover **500** according to an embodiment of the present disclosure. Since the flange portion **510** is formed at the cover **500**, it is possible to easily couple the cover **500** and the pump housing **220** by welding the flange portion **510** and the pump housing **220** or using a screw, etc. Further, since the flange portion **510** is in close contact with the wall of the pump housing **220**, there is an effect of preventing a working fluid from leaking from the inside of the pump housing **220**.
- (40) The sensed component **260** and a portion of the shaft **250** may extend toward the electronic control unit **411** through the pump housing **220**. In this case, the inner circumferential surface of the cover **500** may be machined in a cylindrical shape spaced a predetermined gap from the outer circumference of the shaft **250** such that the portion protruding toward the electronic control unit **411** can be accommodated and rotated.
- (41) Accordingly, even if a working fluid leaking from the pump **240** flows along the shaft **250** rotating in the pump housing **220**, it is possible to prevent the working fluid from leaking out of the electronic control unit **411**.
- (42) As described above, according to this embodiment, the pump housing layout for a brake system according to the present disclosure has an effect of preventing a working fluid from leaking to the electronic control unit from the pump housing by connecting the suction channel of the pump

to the oil reservoir.

(43) Although exemplary embodiments of the present disclosure have been described for illustrative purposes, those skilled in the art will appreciate that various modifications, additions, and substitutions are possible, without departing from the idea and scope of the claimed invention. Therefore, exemplary embodiments of the present disclosure have been described for the sake of brevity and clarity. The scope of the technical idea of the present embodiments is not limited by the illustrations. Accordingly, one of ordinary skill would understand the scope of the claimed invention is not to be limited by the above explicitly described embodiments but by the claims and equivalents thereof.

Claims

1. A pump system for a brake system, comprising: an oil reservoir; a pump housing having a suction port, a discharge port and an inner space; a motor positioned at a first side of the pump housing; an electronic control unit positioned at a second side of the pump housing opposite to the first side; a pump housed at the pump housing and having (1) a suction channel connected to the suction port of the pump housing and (2) a discharge channel connected to the discharge port of the pump housing; a shaft accommodated at the inner space of the pump housing, coupled to and extending from the motor toward the electronic control unit, and configured to be rotated by the motor, the shaft having a rotating body configured to operate the pump and a first end facing the electronic control unit; a sensed component disposed at the first end of the shaft; a sensor disposed on the electronic control unit, spaced apart and facing the sensed component, and configured to sense rotation of the sensed component; a cover disposed between and spaced apart from the sensed component and the sensor and attached to the pump housing to separate the inner space of the pump housing from the electronic control unit; and a pump-reservoir pipeline configured to connect the suction channel to the oil reservoir, wherein the cover is separated from the electronic control unit and comprises: a sidewall surrounding a portion of the shaft protruding outwardly from an outer surface of the pump housing; a flange connected to and surrounding a bottom portion of the sidewall and attached to the outer surface of the pump housing; and a top wall connected to and surrounded by a top portion of the sidewall and positioned between the sensed component and the sensor.
 2. The pump system of claim 1, wherein the pump comprises a piston in contact with the rotating body and configured to reciprocate.
 3. The pump system of claim 1, wherein the pump comprises a piston in contact with the rotating body and configured to reciprocate when the rotating body is rotated.
 4. The pump system of claim 1, wherein the rotating body comprises a driving gear.
 5. The pump system of claim 4, wherein a distance in a rotational axial direction of the shaft between the suction port and the sensed component is smaller than that between the discharge port and the sensed component.
 6. The pump system of claim 1, wherein the pump comprises a plurality of pumps spaced apart from each other in a direction perpendicular to an axial direction of the shaft.
 7. The pump system of claim 1, wherein the electronic control unit is separated from an inside of the pump housing and positioned externally to the pump housing.
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